

AI-Driven Autonomous Campus Energy Intelligence System (ACEIS)

A Project Report (Project-I) submitted in partial fulfillment of the requirements for the award of degree of

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE & ENGINEERING

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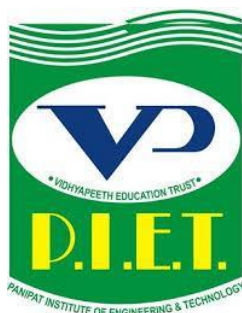
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VI Semester



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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We certify that

- i. The work presented in this project report is an authentic record of our own work under the guidance of our supervisor. It has not been submitted to any other Institute for the award of any other degree or diploma.
- ii. Whenever we have used information (text, data, figure, photograph, chart, analysis, inference, etc.) from other sources, we have given due credit by citing it in the text of the report and providing its details in the references.
- iii. We have followed the guidelines provided by the department for preparing the report.

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Project report title: **AI-Driven Autonomous Campus Energy Intelligence System (ACEIS)**

Semester: VI

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APPROVAL FROM SUPERVISOR

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It is recommended that the report be accepted as fulfilling this part of the requirements for the award of the degree.

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ABSTRACT

With the increase in laboratory and classroom usage and building occupancy, there is an increase in energy consumption in educational institutions. Majority of universities and colleges employ conventional energy management systems that track energy usage but lack intelligent insights in order to optimize their energy consumption. Consequently, inefficient use of energy increases operational cost as well as greenhouse gas emissions.

The project aims to design and develop the proposed solution called Artificial Intelligence Driven Autonomous Campus Energy Intelligence System (ACEIS). ACEIS is designed to monitor the university's electricity consumption and forecast future electricity usage while recommending optimized electricity usage strategies based on the analysis of energy usage history and weather pattern.

Machine learning models such as ARIMA, XGBOOST and Long Short Term Memory (LSTM) will be applied to perform electricity demand forecasting and optimization based on the obtained predictions. The cost-conscious optimization engine will apply linear programming methods to schedule optimal energy usage. The renewable energy simulation will estimate solar energy generation and battery capacity usage while an isolation forest will detect anomalies.

The system provides an interactive dashboard for visualizing energy trends, demand forecasts, optimization results, carbon emission reports, and anomaly alerts. By integrating Artificial Intelligence, forecasting, optimization, and sustainability analysis, ACEIS serves as an intelligent decision-support system that can help institutions reduce energy costs, improve operational efficiency, and promote environmentally responsible energy management.

Keywords: *Artificial Intelligence, Energy Management, ARIMA, XGBOOST, LSTM, Energy Forecasting, Optimization, Anomaly Detection, Smart Campus.*

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
ACEIS	AI-Driven Autonomous Campus Energy Intelligence System
AI	Artificial Intelligence
ML	Machine Learning
LSTM	Long Short-Term Memory
ARIMA	Auto-Regressive Integrated Moving Average
LP	Linear Programming
IoT	Internet of Things
RMSE	Root Mean Square Error
MAE	Mean Absolute Error
API	Application Programming Interface
HVAC	Heating, Ventilation and Air Conditioning
ToU	Time-of-Use
kWh	Kilowatt-hour
CO ₂	Carbon Dioxide
UI	User Interface
SRS	SRS Software Requirement Specification
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
React.js	JavaScript Library for User Interface Development
MongoDB	Mongo Database
Flask	Python-Based Web Framework

CHAPTER 1: INTRODUCTION

1.1 OVERVIEW OF CAMPUS ENERGY MANAGEMENT

Energy utilization in institutions is high due to the numerous laboratory equipment used in classrooms, hostel power, offices, and other places. The use of such systems as energy monitoring is very low because these systems collect data about energy consumption without providing tools to analyze, forecast, and optimize its use. Therefore, institutions experience inefficiency in energy utilization, operational challenges, and environmental issues.

Campus Energy Management aims at monitoring, analyzing, and optimizing energy consumption. The modern energy management systems are capable of forecasting the energy demands, recognizing anomalies in energy utilization patterns, and suggesting efficient ways to manage energy utilization.

1.2 NEED FOR INTELLIGENT ENERGY MANAGEMENT SYSTEMS

The high demand for energy coupled with the increase in energy prices poses a challenge in finding more intelligent ways of managing energy. Traditionally, energy monitoring systems only offer information regarding energy utilization without suggesting how to utilize energy efficiently.

Intelligent energy management systems allow for accurate forecasting, optimize energy consumption, incorporate renewable energy sources, and promote data-based decisions.

1.3 MOTIVATION FOR THE PROJECT

The main motivation behind the development of ACEIS is the issue of inefficiency of energy consumption and rising costs of electricity within educational institutions. While most campuses gather energy information, its analysis in terms of forecasting and optimization is often ignored.

By employing the principles of Artificial Intelligence, Machine Learning and optimization, ACEIS intends to revolutionize energy monitoring into an intelligent decision making tool that will foster energy efficiency, reduce costs and improve sustainable energy management.

1.4 PROJECT GOALS AND OBJECTIVES

The following objectives shall be achieved by means of this project:

- Analysis of campus energy consumption.
- Prediction of future electricity demands using ARIMA, XGBOOST and LSTM algorithms.
- Optimization of energy usage through cost-based scheduling.
- Modeling of renewable energy and batteries usage.
- Determining of carbon dioxide emissions.
- Detecting abnormal energy consumption through the use of anomaly detection techniques.
- Development of interactive dashboards.

1.5 SCOPE OF THE PROJECT

The scope of ACEIS revolves around creating a software platform to effectively manage energy consumption in campuses using intelligent techniques. The system will perform analysis, predictions, optimizations of energy consumption schedule, and give sustainability information through analysis of carbon emissions.

The system is created in a simulated environment without any need for physical deployment of IoT hardware. It is however possible to extend the system in future to incorporate real time smart metering, automation, cloud computing and smart campus applications.

CHAPTER 2: LITERATURE REVIEW

2.1 INTRODUCTION TO LITERATURE REVIEW

Introduction The literature review offers a brief description of all the research works conducted in the field of energy management, forecasting, optimization, and sustainability. In this chapter, an attempt will be made to identify some of the gaps in the previous research carried out in the area of intelligent energy management systems that will drive the research on the ACEIS model. With the advent of artificial intelligence and machine learning, energy management systems have transformed from mere monitoring systems to intelligent ones.

2.2 TRADITIONAL ENERGY MONITORING SYSTEMS

Energy monitoring systems have conventionally been created to measure and provide an indication of power use in the form of statistics. While energy monitoring systems allow for energy usage tracking by organizations, the level of analytics available with such monitoring systems is very minimal. The absence of predictive modeling and anomaly identification leaves organizations with no choice but to act reactively.

2.3 SMART ENERGY MANAGEMENT SYSTEMS

Smart energy management systems integrate information gathering, analysis, and automation to enhance energy efficiency. The smart energy systems use sensor devices, smart meters, and software platforms to gather energy consumption details in real-time. According to scientific literature, smart energy systems are capable of minimizing energy wastage and promoting energy sustainability.

2.4 MACHINE LEARNING IN ENERGY FORECASTING

- Forecasting of future energy demand is becoming one of the main applications of Machine Learning. Models based on analyzing past data patterns allow estimating the future needs in electrical power.
- The most common types of forecasts are ARIMA, XGBOOST, and Long Short-Term Memory (LSTM) models. They allow organizations to plan their energy use better and decrease the risks related to energy management.

2.5 OPTIMIZATION TECHNIQUES IN ENERGY MANAGEMENT

Optimization algorithms are extensively employed for minimizing electricity expenses and optimizing the use of resources. Tools like Linear Programming can be used in determining optimal schedules of energy usage, taking into consideration operational restrictions. Research

studies have revealed that the use of optimization algorithms leads to improved energy efficiency without impacting routine operations.

2.6 RENEWABLE ENERGY INTEGRATION

The incorporation of renewable energy sources like solar energy has now emerged as a significant component of modern energy management systems. The use of renewable energy can minimize our dependency on conventional energy sources and minimize our carbon footprint. Some research works indicate the advantages that emerge from using the combination of renewable energy and batteries for savings.

2.7 ANOMALY DETECTION IN ENERGY SYSTEMS

These anomaly detection techniques are used to identify the anomalies in energy consumption, which might indicate malfunctioning in the device, energy loss, or any misuse. Machine Learning algorithms such as Isolation Forest have proven to work effectively in detecting the anomalies in the dataset concerning energy consumption.

2.8 RESEARCH GAPS IDENTIFIED

Even with notable developments in energy management techniques, various issues remain unresolved:

- Lack of adequate incorporation of forecasting, optimization, renewable energy assessment, and anomaly detection into one system.
- Insufficiency of AI-enabled decision-making support in education institutions.
- Lack of any energy intelligence systems aimed specifically at campuses.
- Neglect of carbon emission assessment.
- Reliance on traditional approaches that utilize manual decision-making processes..

2.9 SUMMARY OF LITERATURE REVIEW

According to the literature review, Artificial Intelligence, Machine Learning, Optimization methods, and Renewable Energy systems have a high degree of potential for managing energy. Nonetheless, the current available tools are more focused on their functionality alone, as opposed to being a combined intelligent energy management system. ACEIS, on the other hand, incorporates energy forecasting, cost-effective optimization, renewable energy generation simulation, carbon emission assessment, anomaly identification, and dashboarding in one system.

CHAPTER 3: PROBLEM STATEMENT AND GAP ANALYSIS

3.1 PROBLEM DEFINITION

Schools use a great deal of electrical energy due to their classrooms, laboratories, hostels, offices, and air conditioning systems. The majority of schools operate using traditional monitoring systems which monitor energy use but do not provide any intelligent recommendations for forecasting and optimization.

As a result of the lack of intelligent monitoring systems, schools end up using energy inefficiently, spending more money on power bills, having more expensive peak demands, and causing extra emissions of carbon dioxide. Hence, it is important to develop an intelligent energy consumption monitoring and forecasting system.

3.2 EXISTING SYSTEM OVERVIEW

Current Energy Management Systems are concerned with collecting data and measuring energy consumption. The current smart meters and monitoring devices measure energy consumption and show basic graphical analysis for the manager of facilities.

While it is true that current EM systems aid the process of energy management, they are not equipped with more sophisticated abilities like forecasting, optimizing costs, incorporating renewable energy, and detecting anomalies. Decision-making on energy management still depends on manual processes.

3.3 LIMITATIONS OF EXISTING SYSTEMS

Key deficiencies in the current system include:

1. Ineffective prediction of future energy requirements.
2. Ineffective optimization to minimize electricity bills.
3. Little consideration is given to integrating renewable sources of energy and battery storage systems.
4. Unable to notice abnormalities in energy usage.
5. Carbon footprint is not considered.
6. Energy management approaches are reactive rather than proactive.

3.4 PROPOSED SOLUTION

This proposed ACEIS system can help address these limitations due to the use of an intelligent approach.

The ACEIS system incorporates Artificial Intelligence (AI) technologies using Machine Learning (ML) models like ARIMA, XGBOOST, and LSTM to predict future energy consumption taking into account energy consumption history as well as weather history. Furthermore, there is also an optimization component of this system that uses linear programming to generate optimal energy consumption schedules. Some other components include simulations related to renewable energy generation, emissions calculation, and detection of anomalies using the isolation forest method.

There will be an interactive dashboard designed to display all these parameters.

3.5 ADVANTAGES OF THE PROPOSED SYSTEM

- Prediction of future energy requirements by AI-based predictive models.
 - Lowering of electricity costs by means of efficient energy scheduling.
 - Battery utilization in renewable energy systems.
 - Early detection of any abnormal energy consumption pattern.
 - Measurement of carbon footprint and sustainability report.
 - One centralized platform for data analytics and decision-making.
- Scalable design that can be applied to university settings or smart campuses.

CHAPTER 4: PROPOSED SYSTEM AND FEATURES

4.1 OVERVIEW OF THE PROPOSED SYSTEM

An AI-Driven Autonomous Campus Energy Intelligence System (ACEIS) is defined as an intelligent energy management system that would help institutions track, forecast, and optimize their energy consumption by using artificial intelligence and machine learning to achieve better energy efficiency through optimized energy usage.

ACEIS uses the historical data of energy usage and weather trends to forecast future trends in energy consumption; suggests an optimal scheduling of energy consumption; detects any irregularity in energy consumption; and also predicts the carbon footprints resulting from this usage.

4.2 SYSTEM OBJECTIVES

The key objectives of the proposed system are as follows:

Monitoring and analyzing the energy usage in the college.

- Predicting the future energy requirements through ARIMA, XGBOOST, and LSTM models.
- Optimizing energy usage to save money.
- Modeling renewable energy generation and battery storage.
- Calculating carbon footprint and the effect on the environment.
- Detecting anomalies in the energy usage pattern through Machine Learning.
- Presenting the output results on an easy-to-use dashboard.

4.3 FUNCTIONAL FEATURES OF THE SYSTEM

Energy Consumption Monitoring

The system monitors energy consumption statistics from various facilities across the campuses, including academic facilities, labs, hostels, administrative facilities, and HVAC facilities.

Demand Forecasting

The system uses forecasting models such as ARIMA, XGBOOST, and LSTM to forecast electricity consumption based on previous energy consumption and meteorological data.

Optimization Algorithm

The system uses Linear Programming to create energy scheduling recommendations. It takes into account the constraints and electricity tariffs during energy utilization to reduce energy costs.

Renewable Energy Scheduling Module

This component simulates the production and utilization of renewable energy such as solar energy to compare it with the energy produced by traditional grids.

Carbon Emission Analysis

The system calculates carbon emissions from the electricity consumed and produces carbon emission reports. It calculates reduction in carbon emissions from optimization and renewable energy use.

Anomaly Detection System

Isolation Forest model is used to detect any anomaly in energy consumption statistics. Anomalies such as excessive loads, equipment malfunctions, energy leakages, or abuse of energy can be detected.

Dashboard and Visualization

This system offers an interactive dashboard, which enables users to visualize energy consumption, forecasts, optimization suggestions, carbon emission information, and anomalies in an easy-to-use format.

4.4 NON-FUNCTIONAL REQUIREMENTS

- Scalability: The design makes it possible to scale up and accommodate many departments and larger campuses.
- Security: Reliable forecasts and optimizations are provided by validated models.
- User-friendliness: The dashboard makes it easy for users to navigate.
- Performance: It is capable of handling large amounts of energy-related data.
- Maintainability: Updates will not be difficult because of the modular design.

4.5 ADVANTAGES OF THE PROPOSED SYSTEM

- Increased energy efficiency using advanced analytical capabilities and forecasts.
- Cost savings in terms of electricity usage through optimized scheduling.
- Enables planning and development in sustainable energy practices.
- Enables early detection of abnormal energy usage.
- Helps to detect any anomaly in energy usage.
- Conducts carbon footprint analysis for environmental benefits.

4.6 LIMITATIONS

- Forecasting is dependent on historical data.
- Renewables forecast is simulated based on certain pre-defined scenarios.
- Physical installation of IoT smart meters is currently non-existent in this prototype system.
- Real-time automatic control actions are future work.

CHAPTER 5: SYSTEM ARCHITECTURE AND WORKFLOW

5.1 SYSTEM ARCHITECTURE OVERVIEW

The Artificial Intelligence Driven Autonomous Campus Energy Intelligence System (ACEIS) is a modular system that incorporates data acquisition, forecasting, optimization, sustainability analysis, and visualizations in a cohesive system. The ACEIS system processes data related to energy consumption and weather patterns in order to give intelligent suggestions related to energy management.

There are many different components in the ACEIS system including the Smart Monitoring Layer, Forecasting Engine, Cost-Aware Optimization Engine, Renewable Energy Module, Carbon Emission Intelligence Module, Anomaly Detection System, and Dashboard Visualization Layer. These modules work together to make decisions while making the system intelligent.

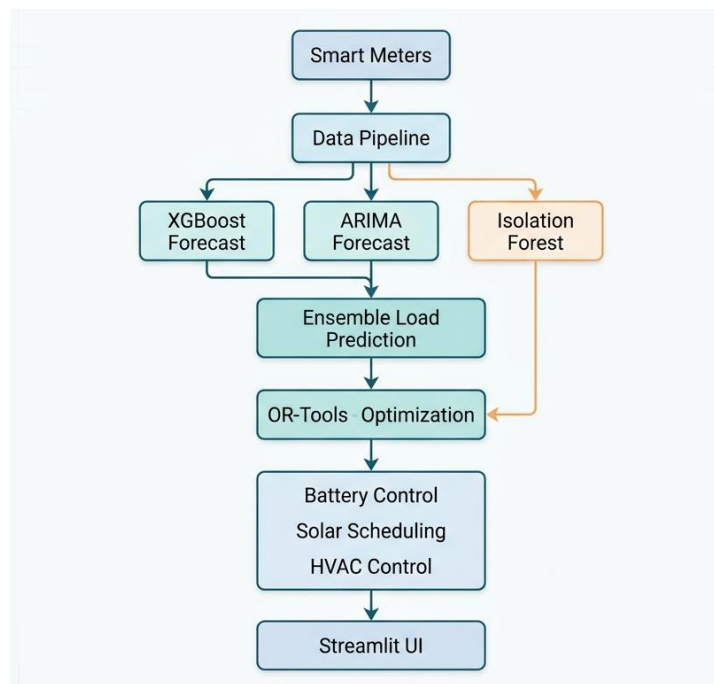


Figure 5.1 – ACEIS System Architecture

5.2 SMART MONITORING LAYER

Smart Monitoring Layer acts as the data collecting layer of the system. This layer collects energy consumption data from different facilities located on campus such as academic buildings, labs, hostels, offices, and heating and cooling systems.

Apart from energy consumption data, weather data including temperature and other environmental factors are also collected in order to increase the accuracy of prediction. The data collected is then processed and transformed into structured data sets.

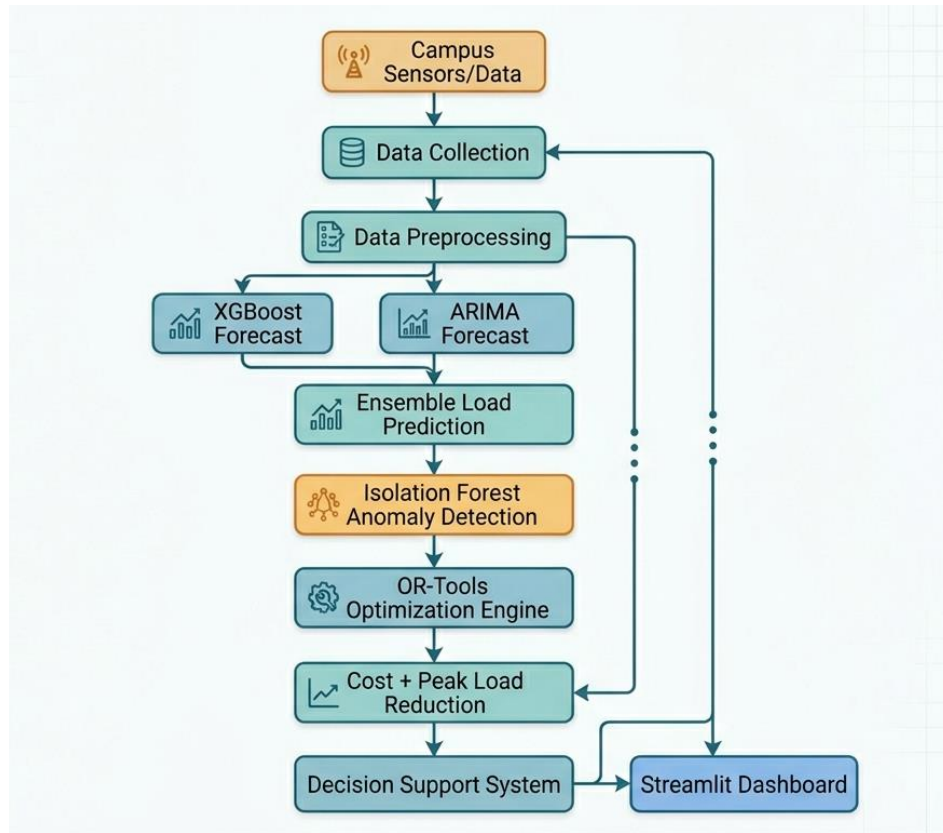


Figure 5.2 – ACEIS System Workflow Diagram

5.3 FORECASTING ENGINE

- The Forecasting Engine is tasked with the job of predicting future power demands. Information based on previous consumption data and weather factors serves as inputs for the forecast process.
- Two types of forecast methodologies are employed by the system:
 - ARIMA (Auto-Regressive Integrated Moving Average)
 - XGBOOST
 - LSTM(in development)

The effectiveness of each of the two forecasting methods is assessed based on metrics like RMSE and MAE.

5.4 COST-AWARE OPTIMIZATION ENGINE

Cost-Aware Optimization Engine focuses on the reduction of energy costs through the creation of effective energy scheduling suggestions. Optimization procedures take into account operational restrictions and electricity tariffs.

Techniques of Linear Programming are applied in order to cut down costs while maintaining smooth functioning of the necessary processes. Energy optimization is helpful in cutting down peak load and improving energy efficiency.

5.5 SIMULATION OF RENEWABLE ENERGY AND BATTERY USE

There is a module which allows simulating the use of renewable energy as well as battery utilization. The amount of solar energy received depends on weather characteristics and assumed efficiency coefficients.

Battery utilization takes into account conditions of charge/discharge process, state-of-charge parameters, and forecasted energy demand. The module provides information about the most efficient combination of solar energy and electricity from the grid.

5.6 CARBON EMISSION INTELLIGENCE MODULE

Purpose of Carbon Emissions Intelligence Module:

This module is used to evaluate carbon footprint due to the usage of energy on campus. The carbon emission calculation uses the emission factor for electricity consumption.

Reports from the module will contain data on carbon footprints and emissions savings percent, among others.

5.7 ANOMALY DETECTION SYSTEM

Anomalies in the energy usage pattern are identified by the Anomaly Detection System, which indicates a possible malfunction of the system, energy leaks, or even theft.

The Isolation Forest Machine Learning method is used for detecting anomalies in energy usage. Upon detection of anomalies, alerts will be issued and visible on the dashboard.

5.8 DASHBOARD AND VISUALIZATION LAYER

The Dashboard Layer serves as the user interface for the whole process. All the trends regarding energy consumption, predictions made by the forecast layer, energy recommendations, data analysis on renewable energy sources and batteries, information on carbon emissions, and anomalies can be shown through dynamic charts and graphs.

The dashboard allows facility managers to access information regarding energy consumption and act accordingly.

5.9 WORKFLOW OF THE SYSTEM

The above presents the overall process involved in the system known as ACEIS:

- Step 1: Data collection concerning energy use and weather data.
- Step 2: Data preprocessing and cleansing.
- Step 3: Data classification based on different sectors of the energy use.
- Step 4: Use of ARIMA, XGBOOST and LSTM models to forecast future energy use.
- Step 5: Evaluation of models and selection of the best one.
- Step 6: Generation of energy scheduling advice via the optimization process.
- Step 7: Renewable energy generation and battery utilization analysis.
- Step 8: Estimation of carbon emissions and preparing sustainability report.
- Step 9: Detection of any anomalies in the energy use data.

CHAPTER 6: TECHNOLOGY STACK AND TOOLS

6.1 OVERVIEW OF TECHNOLOGY STACK

For the effective design and implementation of the AI-Driven Autonomous Campus Energy Intelligence System (ACEIS), there is a need for integration of several technologies such as machine learning, data analysis, optimization, anomaly detection, and visual interaction. The purpose of ACEIS is to be able to make accurate predictions regarding campus energy demand, detecting abnormalities, energy optimization, and making intelligent decisions.

The selected technology stack emphasizes simplicity, scalability, and ease of deployment, yet at the same time allows for AI and ML operations.

6.2 FRONTEND TECHNOLOGIES

6.2.1 Streamlit

Streamlit is used as the primary framework to design a user-friendly dashboard for ACEIS. Using this framework makes it possible to create an interactive dashboard that displays the outcomes of predictions, optimization, energy consumption, battery consumption, solar power generation, and anomalies detection.

Main features include:

- User interface
- Real-Time forecasts visualization
- Interactive simulation control
- Cost analysis dashboard
- Optimization process monitoring
- Anomaly detection notifications

6.3 BACKEND TECHNOLOGIES

6.3.1 Python

Python is the programming language at the core of the ACEIS project. Python can be used in tasks related to data preprocessing, modeling using AI techniques, optimization, anomaly detection, and dashboard creation.

The reason for choosing to use Python in the ACEIS project lies in its many available AI and optimization libraries.

6.4 MACHINE LEARNING AND FORECASTING TECHNOLOGIES

6.4.1 XGBoost

Extreme Gradient Boosting is employed as the machine learning technique for predicting campus energy demand. It uses several variables like temperature, occupancy rate, heating, ventilation, and air conditioning load, solar production, and electricity rates as inputs.

These include:

- Forecasting accuracy
- Fast learning and prediction
- Dealing effectively with non-linear dependencies
- Successful working with tabular data sets

6.4.2 ARIMA Model

ARIMA stands for Auto-Regressive Integrated Moving Average. This model uses statistics to make predictions of future energy requirements from past data and trends.

Examples are:

- Load forecasting
- Trends analysis
- Energy baseline prediction

6.4.3 Ensemble Forecasting

ACEIS utilizes the outputs from the predictions of XGBoost and ARIMA through an ensemble technique for forecasting. Through the use of this forecasting technique, enhanced reliability and forecasting accuracy are achieved through harnessing the power of machine learning and statistics.

6.4.4 Isolation Forest

The Isolation Forest model is deployed for the purpose of detecting anomalies in the energy consumption patterns of the campus. This model detects abnormal energy consumption behavior, energy spikes, HVAC system malfunctions, and abnormal operating modes.

Examples of applications:

- Equipment malfunction detection
- Energy theft detection
- Abnormal load spike detection
- Predictive maintenance

6.5 DATA PROCESSING LIBRARIES

6.5.1 NumPy

NumPy is utilized for performing numeric calculations, linear algebra, and mathematical transformations that are needed during the forecasting and optimization process.

6.5.2 Pandas

Pandas is utilized to clean and pre-process energy datasets and manage structured datasets. Pandas simplify the handling of time series data and structured campus energy datasets.

6.6 OPTIMIZATION TECHNOLOGIES

6.6.1 OR-Tools

Google OR-Tools is deployed to develop an optimization engine for ACEIS. This optimization engine optimizes energy use by minimizing costs and lowering peak demands while using maximum renewable energy.

Optimization engine carries out:

- Grid energy scheduling
- Battery dispatch optimization
- HVAC load optimization
- Solar energy usage
- Peak load reduction
- Minimum cost optimization

6.7 VISUALIZATION AND ANALYTICS TOOLS

6.7.1 Matplotlib

The visualization tool used to produce forecast graphs, optimization diagrams, cost estimation charts, and energy consumption visualization.

6.8 MODEL EVALUATION TOOLS

6.8.1 MAE (Mean Absolute Error)

This metric is used to measure the forecasting error rate by calculating the average difference between the predicted values and the actual values. Smaller MAE values represent better model accuracy.

6.8.2 RMSE (Root Mean Square Error)

Used to measure the forecasting error by giving more weight to bigger differences between the actual and forecasted values. This method gives an all-around view of forecasting performance.

These metrics help to compare both XGBoost and ARIMA forecasting models, allowing one to determine which model suits the best.

6.9 DEVELOPMENT TOOLS

Development tools are used throughout the process include:

- Visual Studio Code (VS Code) - for source coding development and debugging
- Git and Github – for version control and collaboration
- Streamlit – dashboard creation and visualization
- Joblib – model serialization and persistence
- Jupyter Notebook – data exploration and modeling

These development tools help in efficient platform development, testing, implementation, and maintenance.

CHAPTER 7: SOFTWARE REQUIREMENT ANALYSIS (SRS)

7.1 INTRODUCTION TO SRS

The Software Requirements Specification (SRS) provides details on both the functional and non-functional features of the AI-Driven Autonomous Campus Energy Intelligence System (ACEIS). SRS serves as the basis for meeting all the objectives of the system in its development process.

SRS provides a systematic description of the software for ease in development, testing, implementation, and maintenance of the software development project.

7.2 PROBLEM STATEMENT

Designing intelligent software applications for collecting information about campus energy consumption, forecasting electricity demands, optimizing energy usage, calculating carbon emissions, detecting abnormal energy consumption patterns, and making recommendations using an interactive Web-based dashboard.

7.3 FUNCTIONAL REQUIREMENTS

The above-listed capabilities will be included within the scope of the system's features:

- Energy consumption tracking and management.
- Weather information analysis.
- Forecasting demand of energy via ARIMA and LSTM methods.
- Optimization of energy considering costs using linear programming method.
- Renewable energy and battery consumption analysis.
- Carbon emissions calculation.
- Detection of anomalous energy consumption.
- Data visualization dashboard.
- Information storage about energy.

7.4 NON-FUNCTIONAL REQUIREMENTS

- Performance: It must perform forecasting and optimization operations without fail.
- Reliability: The platform must yield reliable and accurate output.
- Scalability: The system must be scalable to meet new developments.
- Security: All data must be protected against any unauthorized intrusion.
- User Interface: The dashboard of the system must have a user-friendly interface.
- Maintainability: The modularity of the system should make future enhancement possible.

7.5 SOFTWARE REQUIREMENTS

Software / Technology	Purpose
Python	Data processing, forecasting, optimization, and backend development
Streamlit	Interactive web-based dashboard development and visualization platform
Pandas	Data manipulation, preprocessing, and dataset management
NumPy	Numerical computations and array operations
XGBoost	Machine learning model used for energy demand forecasting
ARIMA	Statistical energy forecasting
Optimization Engine (Custom Python)	Load scheduling, battery dispatch, HVAC optimization, and energy cost reduction
Isolation Forest	Anomaly detection
Scikit-learn	Model training, preprocessing, scaling, and anomaly detection support
VS Code	Development environment
Git & GitHub	Version control and project management
Postman	API testing

Table 7.1 Software Requirements

7.6 HARDWARE REQUIREMENTS

Component	Minimum Requirement
Processor	Intel Core i3 or higher
RAM	4 GB or higher
Storage	10 GB free disk space
Display	Standard Monitor/Laptop Screen
Internet Connection	Required for APIs and deployment

Table 7.2 Hardware Requirements

7.7 FEASIBILITY STUDY

7.7.1 Technical Feasibility

From a technical point of view, the proposed system can be implemented because it utilizes tested technologies such as Python, Streamlit, MongoDB, and Machine Learning packages. The technologies mentioned above are widely used, easily understandable, and appropriate for building intelligent energy management systems.

7.7.2 Economic Feasibility

It is economically viable since the system uses open source software as well as free resources for development purposes. The installation costs associated with it are lower compared to savings from energy efficiency.

7.7.3 Operational Feasibility

The ACEIS software solution is one that is quite simple to use and which will allow facility management staff to keep track of energy consumption, generate reports, and take smart decisions through the dashboard.

7.7.4 Environmental Feasibility

This is environmentally-friendly because it ensures that there is no wastage of energy and encourages the adoption of green energy through carbon emissions analysis.

CHAPTER 8: RESULTS AND OUTPUT SCREENS

8.1 INTRODUCTION

Results from this chapter are related to the operation of the AI-based Autonomous Campus Energy Intelligence System (ACEIS). The ACEIS incorporates the energy monitoring, energy demand forecasting, energy cost optimization, renewable energy modeling, carbon emissions calculation, and anomalies prediction into one intelligent system.

Output is displayed on the screen which highlights all steps undertaken in the operation of the system, namely energy analysis, forecasting, and optimization. The results obtained from the dashboard provide useful information for making decisions.

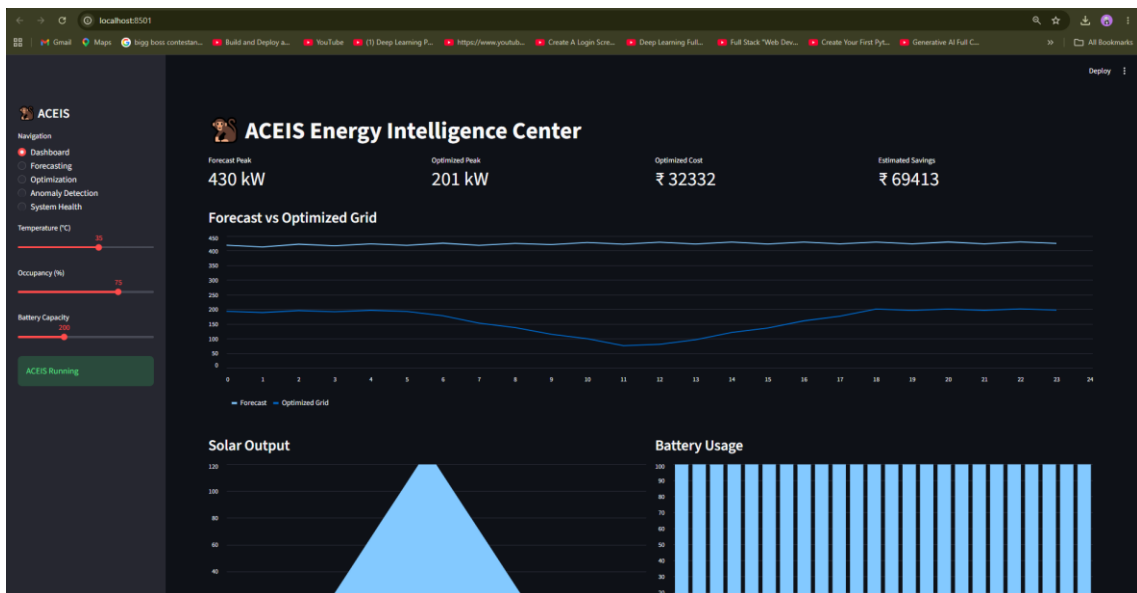


Figure 8.1 – ACEIS Dashboard Home Screen

8.2 DASHBOARD OVERVIEW

ACEIS Dashboard forms the core interface of the software. It provides comprehensive information on issues such as energy usage in the college, future energy needs, optimization, carbon footprint, and anomalies, among others.

It has been developed using Streamlit technology, thus providing a user-friendly interface to the dashboard. Through the use of various charts and graphs, real-time tracking of energy information becomes possible.

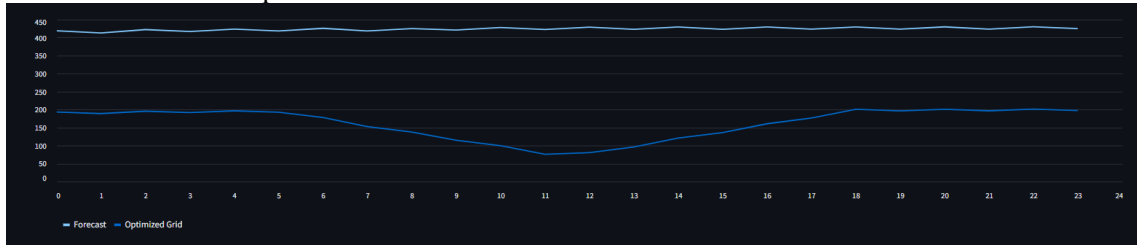


Figure 8.2 – HVAC Analysis Dashboard

8.3 ENERGY CONSUMPTION ANALYSIS

The Energy Consumption Analysis component presents graphical representation of past power usage data from campus premises, including classrooms, labs, hostels, administration offices, and HVAC systems.

Graphical aids make it easier for users to detect consumption trends and time of high-energy consumption. The data obtained through this analysis can prove to be very helpful in decision making.

Multi-Model Forecasting				
Hour	XGBoost	ARIMA	Ensemble	
0	0	473.8708	364.4785	439.1747
1	1	473.7564	351.9326	412.8445
2	2	475.8931	368.6642	422.2786
3	3	475.5343	358.0146	416.7745
4	4	475.1145	372.2363	423.6754
5	5	475.5999	361.2073	418.4036
6	6	476.6136	375.0421	425.8278
7	7	473.532	363.4622	418.4971
8	8	473.4329	376.9385	425.1857
9	9	477.0658	365.0644	421.0651

Figure 8.3 – ARIMA Forecasting Result

8.4 DEMAND FORECASTING RESULTS

In the Forecasting module, two types of models – ARIMA and XGBOOST – are used to forecast future electricity demand. Energy consumption and weather variables have been used to forecast short-term future demand.

According to the forecast results, the model effectively captures energy consumption patterns and accurately forecasts future demand.

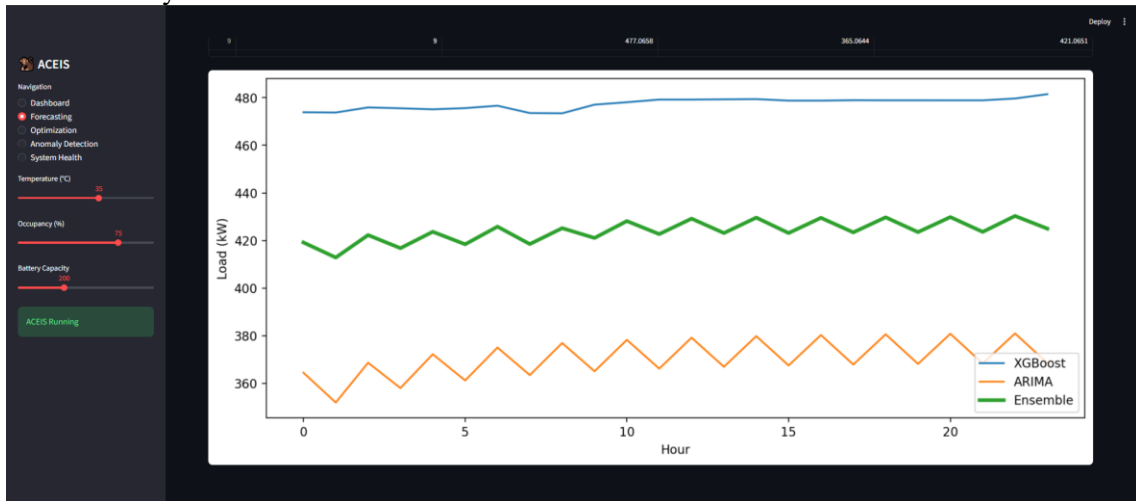


Figure 8.4 – Forecasting Result

8.5 COST OPTIMIZATION RESULTS

The Cost Aware Optimization module creates energy schedules that will minimize electricity costs by ensuring they meet the operational requirements.

From the optimized output, an effective energy balancing is evident. This is achieved through moving some of the loads during the off-peak times thus minimizing the total energy bills.

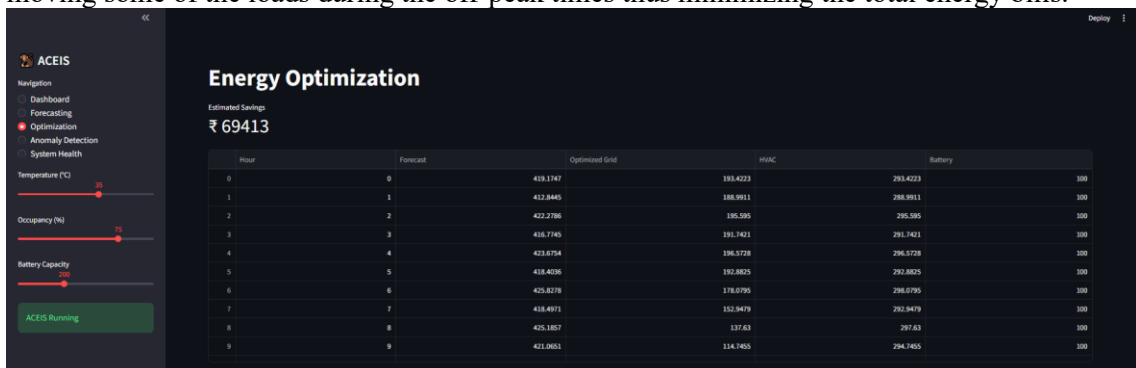


Figure 8.5 – Cost Optimization Dashboard

8.6 RENEWABLE ENERGY SIMULATION RESULTS

Renewable Energy Module calculates solar energy production and battery use based on weather and future demands.

Simulation demonstrates that renewable sources can serve as complementary sources of power in addition to saving money. The module also gives suggestions on when the batteries should be charged and discharged.



Figure 8.6 – Renewable Energy Simulation Output

8.7 CARBON EMISSION ANALYSIS

Calculation of environmental impact based on the energy used at the campus is done using the Carbon Emission Analysis module by applying standard emission rates.

The dashboard displays information on the carbon footprints generated as well as the amount of emission reductions gained through efficient energy use and the incorporation of green energy sources.

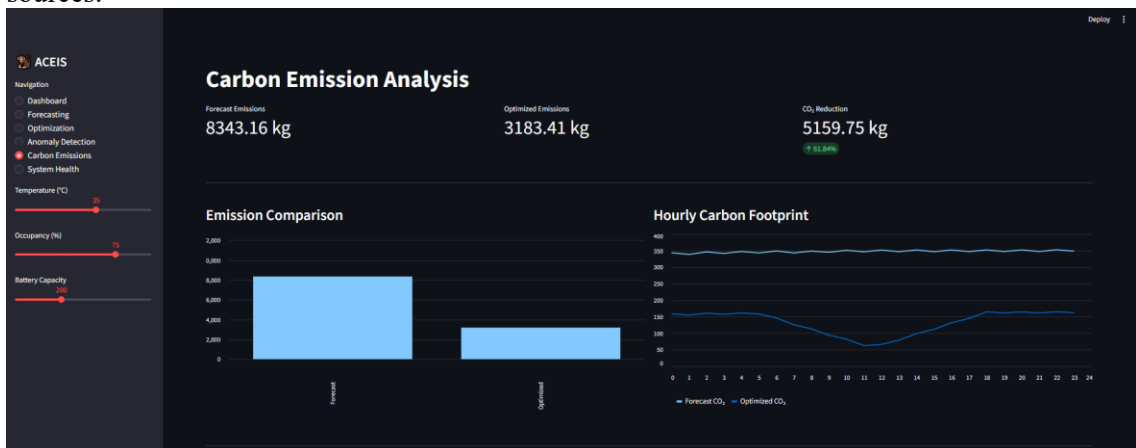


Figure 8.7 – Carbon Emission Analysis Dashboard

8.8 ANOMALY DETECTION RESULTS

The Anomaly Detection Module uses the Isolation Forest algorithm to detect anomalies in energy usage.

This has been successful in detecting anomalies like load surge, energy overusage, and possible equipment failures. These are indicated via dashboard notifications to facilitate prompt action.

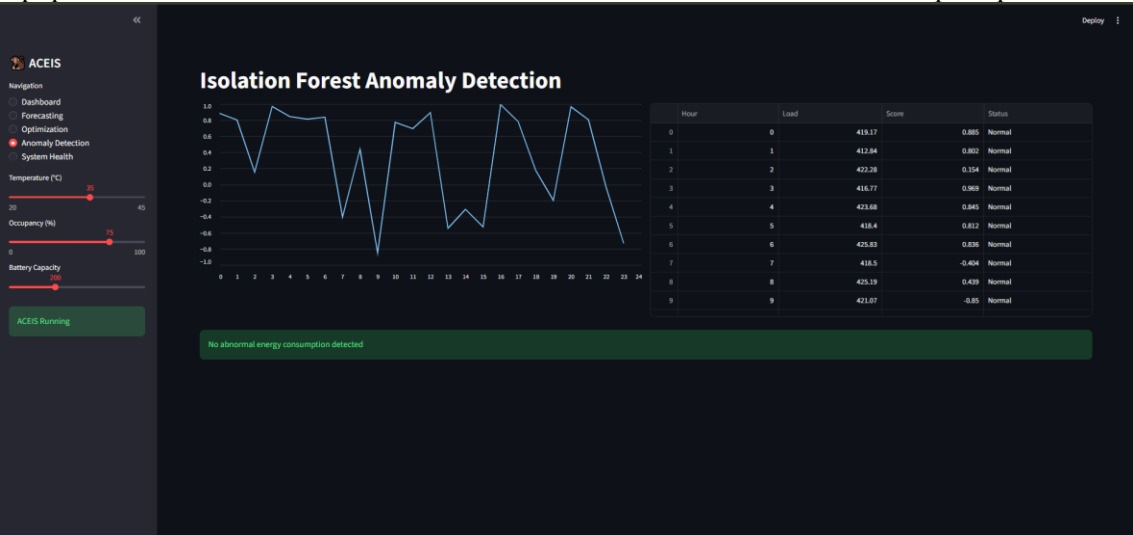


Figure 8.8 – Anomaly Detection Alert Screen

8.9 DISCUSSION OF RESULTS

These outcomes clearly show that ACEIS has the ability to integrate forecasting, optimization, renewable energy considerations, carbon emissions tracking, and anomaly detection in one single system. The forecasting models help produce accurate demand forecasts, while the optimization module helps produce feasible recommendations for minimizing energy expenses. The renewable energy and carbon emissions considerations assist in achieving sustainability goals, and the anomaly detection system enhances reliability. Overall, it can be said that the system has been able to achieve its aim, which is turning traditional energy monitoring into intelligent decision-making for energy management.



Figure 8.9 – Final Integrated ACEIS Dashboard

CHAPTER 9: CONCLUSION

9.1 SUMMARIZATION OF THE PROJECT

The Artificial Intelligence-Based Autonomous Campus Energy Intelligence System (ACEIS) is an intelligent energy management tool designed to increase energy efficiency, minimize operational expenses, and promote sustainable use of energy in educational establishments. Energy monitoring, demand forecasting, cost optimization, renewable energy simulation, carbon emissions analysis, and abnormality detection capabilities are all integrated into the framework.

By applying Machine Learning algorithms such as ARIMA, XGBOOST and LSTM combined with optimization methods and data analytics, ACEIS can be considered a smart decision-making system that makes traditional energy monitoring more efficient.

9.2 ACHIEVEMENT OF OBJECTIVES

The main objectives set at the beginning of the project have been achieved successfully:

- Energy consumption monitoring and analysis.
- Prediction of the upcoming electricity consumption by means of ARIMA, XGBOOST and LSTM methods.
- Optimal energy consumption via cost-based scheduling methods.
- Renewable energy production and battery utilization simulation.
- Estimation of carbon emissions and sustainability assessment.
- Abnormalities detection in energy consumption by applying Machine Learning models.
- Development of an interactive dashboard.

9.3 SIGNIFICANCE OF THE PROJECT

ACEIS system not only provides economic benefits but also helps the environment. The system aids institutions in cutting down on their electricity bills through proper scheduling and effective utilization of resources. Besides this, it also helps in promoting renewable energy and cutting down the carbon footprint.

This project highlights how Artificial Intelligence, Machine Learning, Optimization Techniques, and Data Analytics can be used for practical applications to solve energy management challenges.

9.4 LIMITATIONS OF THE SYSTEM

Though the proposed system performs its expected functions, yet it has the following limitations:

- Accuracy of prediction depends upon data quality and availability.
- Renewable energy estimation is done considering hypothetical conditions.
- No physical deployment of Internet of Things (IoT) smart meter is incorporated in the system.
- No provision of automatic real-time control of electrical system is provided in this project.

The limitations stated above will give some ideas for improvement in future.

9.5 FUTURE SCOPE

The system can be further improved through the following developments:

- Incorporation of real-time IoT smart meters and sensors.
- Provision of automatic energy control.
- Conversion into SaaS Platform.
- Application of Reinforcement Learning techniques.
- Connection with Smart Grid.
- Extending support to multi-campus comparative energy analysis.

Further development can add more value to the system.

9.6 CONCLUDING REMARKS

ACEIS exemplifies how Artificial Intelligence and Machine Learning technologies can be utilized efficiently in addressing contemporary problems related to energy management. The integration of forecasting, optimization, sustainability assessment, and anomalies identification into one tool presents an excellent solution for intelligent energy management on campus grounds.

This successful project can undoubtedly serve as a good example demonstrating that data-driven decision-making can play a crucial role in enhancing energy management and achieving sustainability.

REFERENCES

- [1] G. E. P. Box, G. M. Jenkins, G. C. Reinsel, and G. M. Ljung, *Time Series Analysis: Forecasting and Control*, 5th ed. Hoboken, NJ, USA: Wiley, 2015.
- [2] S. Hochreiter and J. Schmidhuber, "Long Short-Term Memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [3] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [4] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge, U.K.: Cambridge University Press, 2004.
- [5] J. D. Taylor, *Energy Management Handbook*, 8th ed. New York, NY, USA: McGraw-Hill Education, 2017.
- [6] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd ed. New York, NY, USA: Springer, 2009.
- [7] OpenWeather, "Weather Data API Documentation." Available: <https://openweathermap.org/api>
- [8] Bureau of Energy Efficiency (BEE), Government of India, "Energy Efficiency Policies and Guidelines." Available: <https://beeindia.gov.in>
- [9] Ministry of Housing and Urban Affairs, Government of India, "Smart Cities Mission." Available: <https://smartcities.gov.in>
- [10] MongoDB Inc., "MongoDB Documentation." Available: <https://www.mongodb.com/docs>
- [11] React Documentation, "React – A JavaScript Library for Building User Interfaces." Available: <https://react.dev>
- [12] Flask Documentation, "Flask Web Development Framework." Available: <https://flask.palletsprojects.com>
- [13] TensorFlow Documentation, "TensorFlow Machine Learning Framework." Available: <https://www.tensorflow.org>
- [14] Pandas Documentation, "Python Data Analysis Library." Available: <https://pandas.pydata.org>
- [15] NumPy Documentation, "Numerical Computing in Python." Available: <https://numpy.org>

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